

KOWANYAMA, Australia

WMO: 942680

Lat: 15.4817S

Lon: 141.7483E

Elev: 35

StdP: 14.68

Time Zone: 10.00 (E10)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	54.8	57.7	32.3	26.8	73.4	37.5	33.0	74.3	16.4	80.0	14.8	80.6	2.0	130	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	19.7	98.3	72.6	96.7	72.9	95.3	73.5	84.9	88.4	81.7	87.1	80.5	86.6	9.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
84.1	179.2	86.7	80.5	158.5	83.9	79.1	151.2	82.8	49.4	88.0	45.5	87.2	44.2	86.8	98.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
15.6	14.1	12.8	DB	49.0	102.0	3.6	1.6	46.5	103.2	44.3	104.2	42.3	105.1	39.7	106.3
			WB	44.2	91.6	2.4	5.8	42.4	95.8	41.0	99.2	39.6	102.5	37.9	106.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.6	83.3	83.1	82.8	81.3	77.8	74.4	73.7	75.3	80.2	84.3	86.1	85.6
	DBStd	5.11	2.57	2.35	2.09	2.19	3.46	3.85	3.85	3.35	3.01	2.43	2.29	2.74
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	2	0	0	0	0	0	1	1	0	0	0	0	0
	CDD50	11178	1033	926	1016	939	860	732	733	784	907	1064	1082	1102
	CDD65	5705	568	506	551	489	395	283	269	319	457	599	632	637
	CDH74	60770	6160	5344	5681	4956	3746	2516	2597	3139	4598	6667	7661	7705
	CDH80	27028	2433	2016	2234	2151	1577	922	950	1348	2355	3419	3886	3737
(14) Wind	WSAvg	5.9	5.7	5.4	5.1	5.2	5.5	5.9	5.9	6.1	6.4	6.8	6.8	6.1
(15) Precipitation	PrecAvg	51.1	12.7	15.4	9.2	2.1	0.2	0.1	0.1	0.0	0.1	0.7	2.5	6.6
	PrecMax	78.7	31.1	33.4	27.0	10.3	2.7	0.9	0.6	0.6	1.3	11.2	10.0	17.2
	PrecMin	22.2	2.4	1.7	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.7
	PrecStd	14.3	7.0	6.9	5.8	2.4	0.4	0.2	0.1	0.2	0.3	1.8	2.4	3.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.0	94.4	94.0	93.9	93.2	90.5	90.9	93.5	97.4	99.9	99.9	100.3
		MCWB	76.2	78.8	78.0	75.5	72.7	70.3	69.7	70.3	69.8	70.9	71.4	71.9
	2%	DB	93.5	92.6	92.3	92.3	91.2	88.8	88.8	91.1	95.2	97.8	98.1	97.5
		MCWB	77.7	79.0	77.5	75.0	72.2	70.0	68.8	68.3	68.9	71.2	72.7	74.9
	5%	DB	91.8	91.3	91.0	90.9	89.6	87.3	87.3	89.1	93.5	96.0	96.4	95.6
		MCWB	78.2	78.8	77.4	74.7	72.1	69.2	67.9	67.0	68.6	71.4	73.6	75.6
	10%	DB	90.1	89.7	89.7	89.7	88.0	85.6	85.6	87.3	91.6	94.2	94.8	93.6
		MCWB	78.3	78.9	77.2	74.2	71.5	68.4	66.9	66.0	68.3	71.4	74.3	76.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	85.3	84.3	83.3	88.5	86.8	77.4	78.4	80.6	80.9	86.7	89.3	86.2
		MCDB	87.5	90.9	88.3	88.7	87.8	79.7	84.4	83.0	86.5	87.8	93.1	88.1
	2%	WB	81.2	82.2	81.0	79.5	78.4	72.9	72.9	72.0	74.8	77.9	80.7	80.7
		MCDB	87.2	87.8	86.9	85.6	86.0	83.3	82.9	84.1	84.3	84.6	88.0	87.9
	5%	WB	80.2	80.9	80.1	78.1	75.6	71.7	71.0	70.2	73.3	76.4	79.0	79.8
		MCDB	86.4	86.4	85.9	84.5	82.2	82.7	81.9	81.8	83.2	84.3	86.6	86.9
	10%	WB	79.6	80.1	79.2	77.1	74.1	70.5	69.2	68.9	72.0	75.4	77.9	79.1
		MCDB	85.5	85.4	84.9	83.8	81.5	81.3	80.2	80.3	81.9	84.1	85.9	85.9
(35) Mean Daily Temperature Range	5% DB	MDBR	13.3	12.8	14.2	18.2	21.9	23.3	25.4	26.8	26.1	23.0	19.7	16.8
		MCDBR	16.2	15.5	16.2	19.3	22.3	23.8	25.0	27.3	26.8	24.2	21.4	20.0
		MCWBR	5.3	5.2	5.1	6.1	7.7	8.8	9.0	9.5	7.9	8.1	7.9	6.7
	5% WB	MCDBR	13.7	13.4	14.1	16.2	18.7	21.3	21.9	25.0	24.3	21.4	18.5	15.9
		MCWBR	5.6	5.6	5.6	7.3	9.7	9.4	10.9	10.7	8.6	8.8	8.9	6.3
(40) Clear-Sky Solar Irradiance	taub	0.410	0.404	0.387	0.370	0.356	0.344	0.338	0.349	0.394	0.424	0.448	0.437	(40)
	taud	2.509	2.529	2.574	2.597	2.584	2.592	2.594	2.546	2.407	2.343	2.310	2.382	(41)
	Ebn at Noon	297	296	294	289	282	280	285	291	287	286	283	288	(42)
	Edh at Noon	36	35	33	30	29	28	28	31	38	42	44	41	(43)
(44) All-Sky Solar Radiation	RadAvg	1829	1818	1859	1856	1646	1505	1607	1838	2120	2293	2295	2070	(44)
	RadStd	175	236	162	118	73	80	51	45	49	81	123	219	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page